



Computer Architecture
CS 325 - 001
Department of Physics and Computer Science
Medgar Evers College
Exam 1 - Make Up

Instructions:

- Complete on May 25 by 10:00 pm.
- Use the IAS instruction set:

Mnemonic	Opcode	Description
LMA	0A	Transfer contents from MQ to AC
LDM	09	Transfer M(X) to MQ
STA	21	Transfer contents from AC to memory location X
LDA	01	Transfer M(X) to AC
LDN	02	Transfer -M(X) to AC
ALD	03	Transfer M(X) to AC
ALN	04	Transfer - M(X) to AC
BRL	0D	Takes next instruction from left half of M(X)
BRR	0E	Takes next instruction from right half of M(X)
BPL	0F	If AC ≥ 0 , takes next instruction from the left half of M(X)
BPR	10	If AC ≥ 0 , takes next instruction from the right half of M(X)
ADD	05	Add M(X) to AC; put result in AC
AAD	07	Add M(X) to AC; put result in AC
SUB	06	Subtract M(X) from AC; put result in AC
ASB	08	Subtract M(X) from AC; put result in AC
MUL	0B	Multiply M(X) by MQ; put most significant bits of result in AC; least significant in MQ
DIV	0C	Divide AC by M(X); put quotient in MQ and remainder in AC
LSH	14	Multiply AC by 2
RSH	15	Divide AC by 2
STL	12	Transfer AC[28:39] to M(X)[8:19]
STR	13	Transfer AC[28:39] to M(X)[28:39]
HLT	00	Halts

- Cheating of any kind is prohibited and will not be tolerated.
- Violating and/or failing to follow any of the rules will result in an automatic zero (0) for the exam.

Grading

Section	Maximum Points	Points Earned
1	2	
2	7	
Total	9	

1. Write an IAS program, in hexadecimal, of the statement below, given L, M, and N are the addresses C3, 44, and 2D, respectively

$$L = \frac{2N + M}{M - N}$$

It should be 10 instructions.

2. For the previous problem, perform an instruction cycle trace that lists all sequential changes to registers and memory starting from address 000, given the initial values:

Variable	Address	Initial Value
L	C3	591401E104
M	44	0000000032
N	2D	0000000014

It should be 34 steps.